

MD SHAH IMRAN SHOYON

☎ +13522195630 • ✉ shoyon.m@ufl.edu • in LinkedIn • GitHub

• Graduate Research Assistant, University of Florida • shah-imran.github.io • Gainesville, Florida

EDUCATION

Ph.D. in Electrical & Computer Engineering (Ongoing)

January 2025 – Present

University of Florida, Gainesville, Florida

Relevant Coursework: Image Processing and Computer Vision, Fundamental Machine Learning, Semiconductor Packaging, Physical Assurance and Inspection of Electronics, Reconfigurable Computing, Computational Photography.

B.Sc. in Electrical & Electronic Engineering

July 2020

Shahjalal University of Science and Technology, Sylhet, Bangladesh

RESEARCH INTEREST

- Computer Vision, Large Language Models, Vision Language Models, Multi-modal Machine Learning Models

PROFESSIONAL EXPERIENCE

PhD in ECE (Ongoing), University of Florida, GNV, FL, USA

Jan 2025 – Present

Graduate Research Assistant

- Conducted research across physical assurance, counterfeit IC detection, X-ray inspection, multimodal AI, and 3D medical image segmentation, including dataset development, model training, and inspection workflows.
- Researching agentic security, with a focus on identifying and evaluating agentic-like attack surfaces, multi-stage attack cascades, and security risks in LLM-based systems.
- Developing neurosymbolic AI (NeSy) approaches for hardware verification of Systems-on-Chip (SoCs), combining LLM-based reasoning with symbolic constraints, verification history, and traceable validation workflows.

HowardMiller, Zeeland, USA

Oct 2021 – Dec 2024

Full Stack Developer (Remote)

- AI strategy development with Deep Learning for optimizing budget distribution in PPC campaigns (Tensorflow, Keras, Pandas, Scikit-Learn).
- Design and Develop new features, optimize the old ones, and integrate third-party API solutions (Flask, Django, Celery, Redis, Postgres, Linux, Unicorn, Apache, AWS, Reactjs, API).

Gmonster.co, Balatonalmádi, Hungary

Sep 2018 – Oct 2021

Project Lead (Remote)

- Responsible for overseeing the whole development pipeline.
- Designed and Developed the whole product (Mass Email Marketing App) from scratch and the subscription system (Python, PyQt5, Database, Flask, JavaScript, Linux Server).

EnableGeek, Dhaka, Bangladesh

Jun 2022 – Dec 2024

Founder

- Designed SEO pipeline, growing traffic to 500+ daily clicks within a year.
- Managed content and tech teams; published two books on Python and Java on Amazon.

PUBLICATION

Accepted

- Al, Zaman, M., Jawad, A., Santamaria-Pang, A., Lee, H. H., Tarapov, I., See, K., Imran, M. S., Roy, A., Fallah, Y. P., Asadizanjani, N., & Forghani, R. (2025). **WaveFormer: A 3D Transformer with Wavelet-Driven Feature Representation for Efficient Medical Image Segmentation.** https://doi.org/10.1007/978-3-032-04965-0_64 (MICCAI 2025)
- Shah, M., Shajib Ghosh, Istiaq Firoz Shiam, Patrick Craig, Antika Roy, Pratyush Shukla, Varun Sai Vemuri, Hafez Bahrami, and Navid Asadizanjani (2026) **GUIDE-X: VLM-LLM-based AI assistant for artifact**

detection and operator guidance in x-ray inspection. Metrology, Inspection, and Process Control XL, 34. <https://doi.org/10.1117/12.3090548>

- Shajib Ghosh, Patrick Craig, Antika Roy, Md Shah Imran Shovon, Varun Sai Vemuri, John Allgair, Hafez Bahrami, and Navid Asadizanjani (2026) **AURA-PIE: an AI-enabled understanding and retrieval-augmented assistant for physical inspection of electronics** Metrology, Inspection, and Process Control XL, p. 127, Apr. 2026, doi: <https://doi.org/10.1117/12.3090467>.
- Imran Shovon, M. S., Ghosh, S., Craig, P. J., Pai, C.-Y., Huang, C.-C., & Asadizanjani, N. (2025). **PIN-CLASS: Pinhole-Centric Counterfeit Classification of IC Packages.** 2025 IEEE Physical Assurance and Inspection of Electronics (PAINE), 1–7. <https://doi.org/10.1109/paine66113.2025.11320226>
- Shovon, M. S. I., Ghosh, S., Craig, P. J., Huang, C.-C., Pai, C.-Y., & Asadizanjani, N. (2025). **A Novel Framework for Identifying Counterfeit ICs via Pinhole Evaluation.** 2025 IEEE Research and Applications of Photonics in Defense Conference (RAPID), 1–2. <https://doi.org/10.1109/rapid64712.2025.11151374>.
- Craig, P., Roy, A., Ghosh, S., Shah, M., Varshney, N., & Navid Asadizanjani. (2025). **Deep Learning-Driven X-Ray Analysis for High-Throughput Manufacturing.** 1–2. <https://doi.org/10.1109/rapid64712.2025.11151434>.
- Shah, M. (2024). **Evaluating YOLO Object Detection Models Under Adverse Lighting Conditions.** Bulletin of the American Physical Society; American Physical Society. <https://meetings.aps.org/Meeting/MAR24/Session/NN00.63>
- Shah, M. (2024). **Evaluating and Enhancing Image Caption Generation: A Comparative Study of LSTM and GRU Models with Object and Feature Recognition Strategies.** Bulletin of the American Physical Society; American Physical Society. <https://meetings.aps.org/Meeting/MAR24/Session/FF00.50>

PROJECTS HIGHLIGHTS

Agentic Security for LLM-Based Systems | Research Project 2026 – Present

- Investigating agentic-like attacks that exploit fabricated or manipulated interaction surfaces to influence LLM-driven decision making.
- Designing and evaluating multi-vector, multi-stage attack scenarios to study cascading security failures in agentic systems.
- Developing benchmark-style evaluation pipelines for measuring attack success, robustness, and failure propagation across system components.

Neurosymbolic AI for SoC Hardware Verification | Research Project 2026 – Present

- Developing a NeSy verification framework that combines LLM reasoning with symbolic constraints and hardware verification knowledge.
- Integrating traceable verification history and structured evidence to improve grounding, explainability, and reuse of prior verification results.
- Exploring automated reasoning workflows for SoC security verification using LLM-assisted analysis with formal and symbolic validation.

Physical Inspection of Electronics: Counterfeit Detection | Research Project 2025 – Present

- Designed a pinhole-image acquisition and preprocessing pipeline for IC packages, and automated defect ROI extraction.
- Trained and evaluated CNN/transformer classifiers to distinguish authentic vs. counterfeit parts, achieving good precision/recall on an in-house dataset.
- Built a multimodal optical-image dataset and VLM training setup to extend classification and explanation beyond pure vision models.

LLaMA Fine-Tuning with LoRA & QLoRA | Research Project 2025 – Present

- Contributed to building a physical-inspection dataset (collection, labeling standards, QC pipeline).
- Set up reproducible training environments with Pipenv and Pyenv on the lab server
- Assisted in distributed training with torchrun and Refactoring & optimizing the codebase (modularization, speedups, clearer configs).
- Added pytest coverage and performed code reviews to enforce style and reliability.

Machine Learning | Artificial Intelligence | Computer Vision Projects 2017 – Present

- Analyzing the Consequences of High-Intensity Light YOLOv5 and Strategies for Defending Against Adversarial Attacks (YOLO, Python) [GitHub](#)
- Custom Object Detection and Tracking using Darknet (Darknet-yolo, Python) [GitHub](#), [Youtube](#)
- Dataset Creation for Training Custom YOLO Model (YOLO, Linux Shell Scripting) [Drive](#)
- Auto Vehicle Entry Management for Apartment (OCR, Darknet-yolo, Python) [GitHub](#)
- Live Speed Detection of Vehicle (Darknet-yolo, Python) [GitHub](#), [Youtube](#)
- Classification of Tweets from Twitter (Python, NLTK, Scikit-learn)

Academic Projects

2014 – 2020

Collaborated as a "Research Assistant" on three Official Projects supported by the UGC (University Grants Commission), leading the last two initiatives as Team Leader.

- Developing An Autonomous Underwater Vehicle & Navigation System (Arduino, Microcontroller)
- Automatic Attendance Taking with RFID (Python, Socket Programming, SQL)
- Developing An UGV and a Human Assistant (Python, Raspberry Pi, Socket Programming, Image Processing, Lidar)

Robotics & Electronics Projects

2014 – 2020

- Line Follower Robot with Automatic Collision Avoidance System
- Maze & Grid Solver Robot
- Remote Controlled Vehicle
- Auto Balancing Bot (Control System Project)
- DX Ball Game With Dot Matrix Display and Arduino (Microprocessor Project)
- Egg Incubator

Software Projects

2017 – Present

- Tradeshow Lead Scanning Website
- Personal Trainer Website - Video
- Lead Mining Website
- Smart Class Updates
- **Scraping Application:** Craigslist Scraper, LinkedIn Scraper
- **Automation Bot:** Mass Email Marketing App(MVC), Auto SMS Sending Bot, Copy Assister App
- **Desktop Application:** Pi Camera Booth Controller with Python, Workout Generator for GYM - Video

RESEARCH CONTRIBUTIONS

Tech Licenses | Datasets

2025 – Present

- DeepICLogo v2 Dataset: IC Optical Image Dataset for Logo Recognition (Ongoing)

PROPOSAL/GRANT WRITING

- Assisted in preparing the NSF MRI grant proposal “**Generative Multimodal Intelligence for Diverse Scanning to Minimize Expert Oversight (GIDE)**”

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C, VHDL, Matlab, Processing, Arduino, JavaScript, Flutter, Dart, Assembly Language (8086)

Libraries: Scikit-learn, TensorFlow, Keras, PyTorch, YOLO, NumPy, SciPy, Pandas, Selenium, Requests, BeautifulSoup4, Scrapy, Matplotlib, Seaborn, Bokeh, Selenium, Requests, BeautifulSoup4, Scrapy, Django, Flask, React

Databases: SQL, Firebase, MongoDB, Pickle

Environments: Anaconda Environment, Jupyter Notebook, Pipenv

Hardware & Reconfigurable Computing: FPGA, VHDL

TRAINING & EXTRA CURRICULAR ACTIVITIES

- Participated in 12 national Robotics competitions and been in Leader positions for most of them. Most of the competition was on Line Follower Robot, Maze and Grid Solver, Robo fights, etc. Also, I attended some Electronics Project fairs as well.

- Became 1st runner-up of Robomania, Esonance at IUT in 2015. It was a combination of Line following, Obstacle Avoidance, and a Remote Controlled Bot with hands to carry objects to designated places on challenging terrain.
- Top Up IT training on Java conducted by Ernst & Young LLP, India under the LICT project, and the program is certified by George Washington University, USA.
- Completed AWS Cloud Practitioner training course in 2020. Topics consisted of Analytics, Compute and Serverless, Containers, Database, Network and Content Delivery, Storage and Management, Monitoring, and Governance.
- Experience in teaching Maths, Physics, and Chemistry to high school students.
- Organizing member at EEE Fest-2018. Organized robotics competition policy and challenges.

LINGUISTIC PROFICIENCY

- English (Fluent)
- Bengali (Native language)